

PROJECT OVERVIEW

Same-Tail Delay Propagation & Cascade-Risk Prediction in Australian Domestic Aviation

A working summary of the project's core idea, structure, and current status — for personal reference while OpenSky historical access is pending.

The Core Question

Most airport congestion visualisation treats each airport as an independent problem: a map with dots, a delay heatmap by hour, a bar chart of average lateness. This project asks a different question:

Core question

Does delay at one Australian airport propagate through the same aircraft's rotation to cause delay elsewhere, later the same day — and can the airports that transmit, absorb, or amplify that delay be identified, visualised, and ultimately anticipated before it happens?

Australia's domestic network runs on a relatively small pool of aircraft completing rapid turnarounds across long distances, with limited alternate-routing flexibility compared to denser networks like Europe or the US. A delay at one airport does not stay local — it is carried forward by the same tail number, hours later, often hundreds of kilometres away. This structural feature of Australian aviation is rarely, if ever, visualised directly.

Why This Is a Genuinely Novel Angle

- Standard congestion dashboards are per-airport and look backward only ("how delayed was this airport today").
- Standard delay-prediction models (a very common academic/capstone project) predict delay from local conditions — weather, traffic density — treating each flight independently.
- This project instead treats the domestic network as a connected system with feedback: delay generated at Airport A becomes an input condition at Airport B, C, D later that day, via the literal physical aircraft doing the rotation.
- The Australian context (long distances, fewer alternate routings, concentrated hub structure) plausibly makes this propagation effect more pronounced than in markets where this kind of analysis is more commonly done.

The line that makes this unique

Not “will this flight be delayed because of the weather,” but “given a delay happening right now at Airport A, how likely is it to show up as delay at Airport B several hours from now — and which conditions make that transmission more or less likely?”

The Two Layers of the Project

Layer 1 — Observed Cascade (descriptive)

Reconstruct real aircraft rotations from historical ADS-B data (same icao24 across multiple legs in a day), establish typical turnaround benchmarks per route/aircraft type, and detect where a delay on one leg visibly carries forward — or is fully absorbed — before the next leg. Visualised as an animated network map: a “delay weather system” showing congestion intensity per airport, with fading trails showing delay that originated elsewhere and arrived via aircraft rotation.

Layer 2 — Propagation Risk (predictive)

Using the patterns found in Layer 1 as historical training examples, estimate the likelihood that a delay observed right now at one airport will propagate to specific downstream airports later the same day. Inputs are framed around transmission, not origin: given a known current delay, does weather or traffic density at the next airport make that delay more likely to carry forward rather than be absorbed by schedule buffer? Visualised as a forward-projected, lower-confidence extension of the same network map — one coherent visual system, not two separate outputs.

Optional secondary view: “recovery asymmetry” — once congestion spikes at an airport, how long until it returns to baseline, and does that differ structurally by airport (e.g. lower traffic density recovering faster, versus fewer spare aircraft recovering slower).

Data Sources

- BITRE (Bureau of Infrastructure and Transport Research Economics) — Australian domestic on-time performance statistics. Public, free, available now, no approval needed. Used for the chronic-congestion baseline.
- OpenSky Network (via the traffic Python library, Trino historical interface) — individual aircraft (icao24) position, altitude, speed, and vertical rate over time. Used to reconstruct actual same-tail rotations. Requires separate historical/Trino access approval — currently pending.
- OpenSky live API — available immediately, no approval wait. Used in the meantime to validate the full code pipeline (querying, leg segmentation, plotting) on live traffic before historical access is granted.
- Bureau of Meteorology (BOM) / METAR aviation weather reports — for the propagation-risk layer’s weather inputs. Public, free.

Project Stages & Current Status

Stage	Focus	Data Source	Status
1	BITRE national baseline — which routes/airports are chronically congested	BITRE	Available now

Stage	Focus	Data Source	Status
2	Pipeline validation — query, leg-segmentation, plotting logic tested end-to-end	OpenSky live API	Available now
3	Observed cascade reconstruction — real same-tail rotations, delay carry-forward	OpenSky historical (Trino)	Blocked: pending approval
4	Propagation-risk model — likelihood of delay transmission given conditions	Trino + BOM/METAR	Design only, for now
5	Unified visualisation — observed + projected cascade as one network map	All of the above	Final assembly

Trino historical access requires manual approval by OpenSky administrators, separate from basic account registration, and is currently the only blocking dependency. Stages 1 and 2 require no further access and can proceed immediately; Stage 4's design and data-source scoping (weather inputs in particular) can also proceed now, ahead of having labelled cascade data to train against.

What a Successful Outcome Looks Like

1. A believable, real multi-leg aircraft rotation reconstructed from public ADS-B data, with at least one clear instance of delay carrying forward from one leg to the next.
2. A network-level visualisation that shows congestion as something that moves through the system over the course of a day, rather than something that sits statically at one airport.
3. A propagation-risk layer that answers a genuinely different question to standard delay-prediction work: not “will this be delayed,” but “will this delay spread, and where.”
4. One coherent visual story for the final report and presentation, combining what happened with what is likely to happen next — rather than two disconnected outputs bolted together.